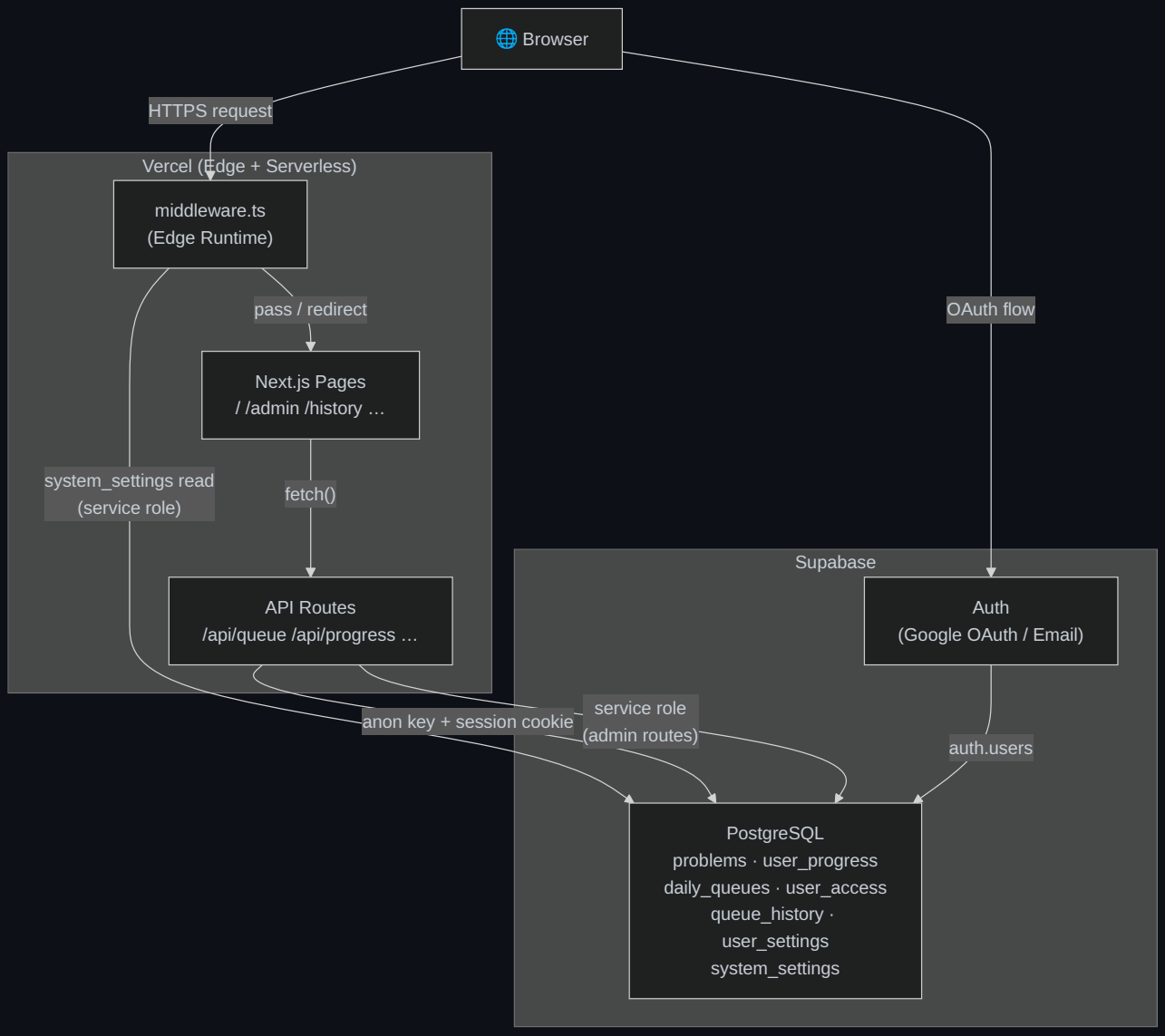
Technical Architecture — DSA Tracker

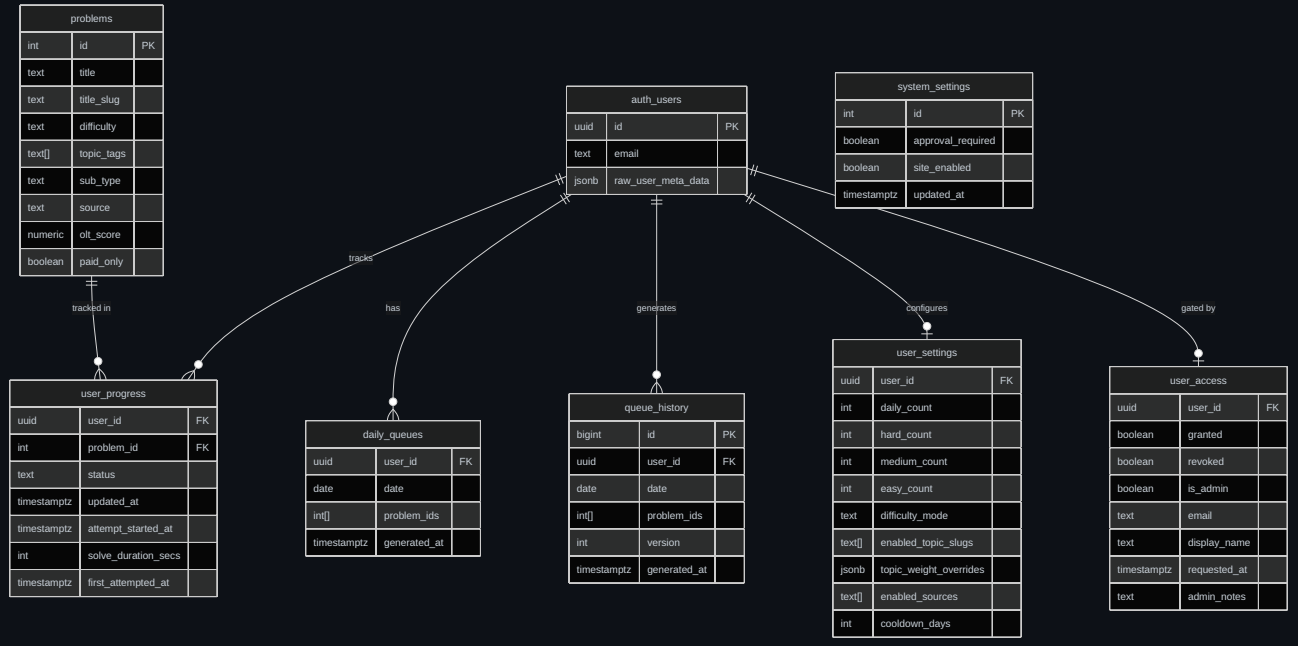
> Last updated: October 2026

Diagrams

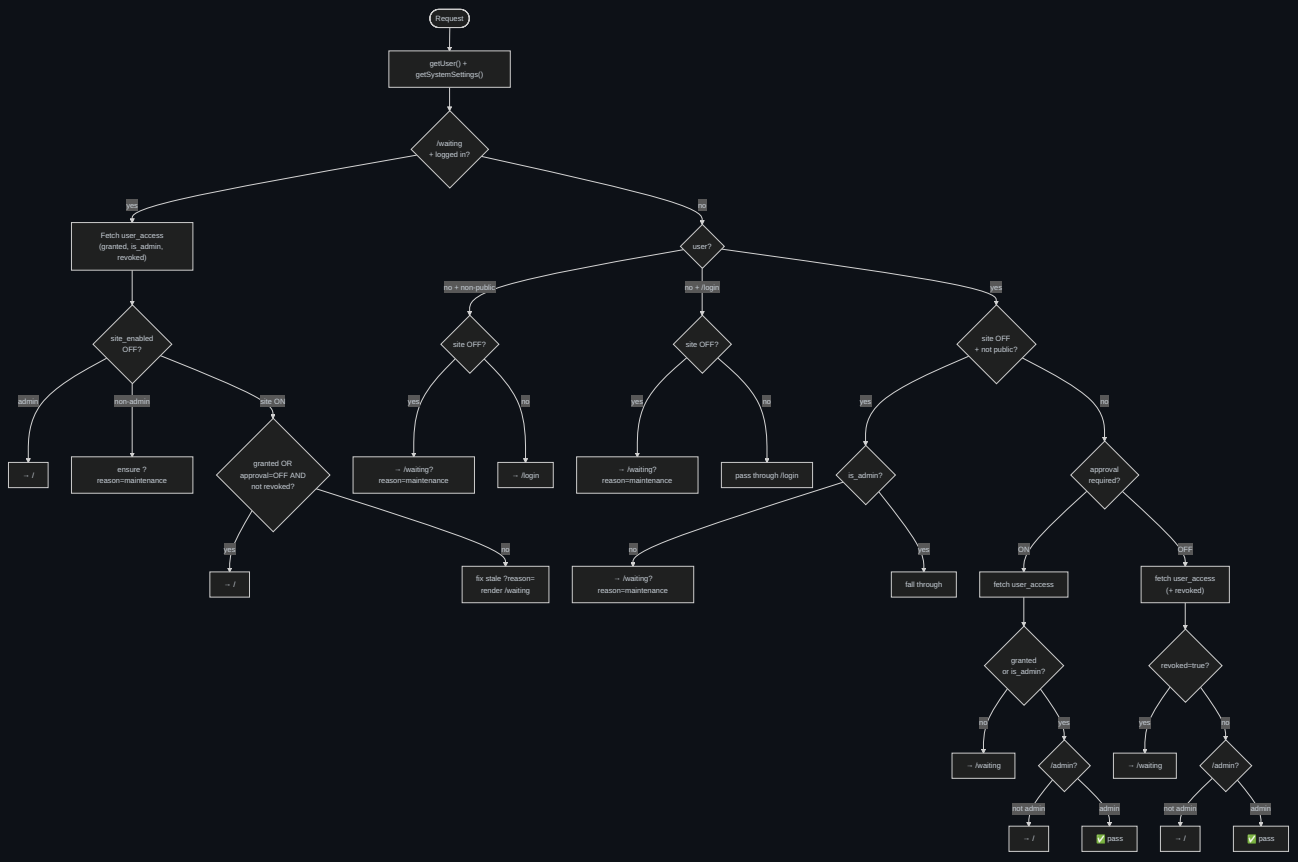
1 — System Architecture



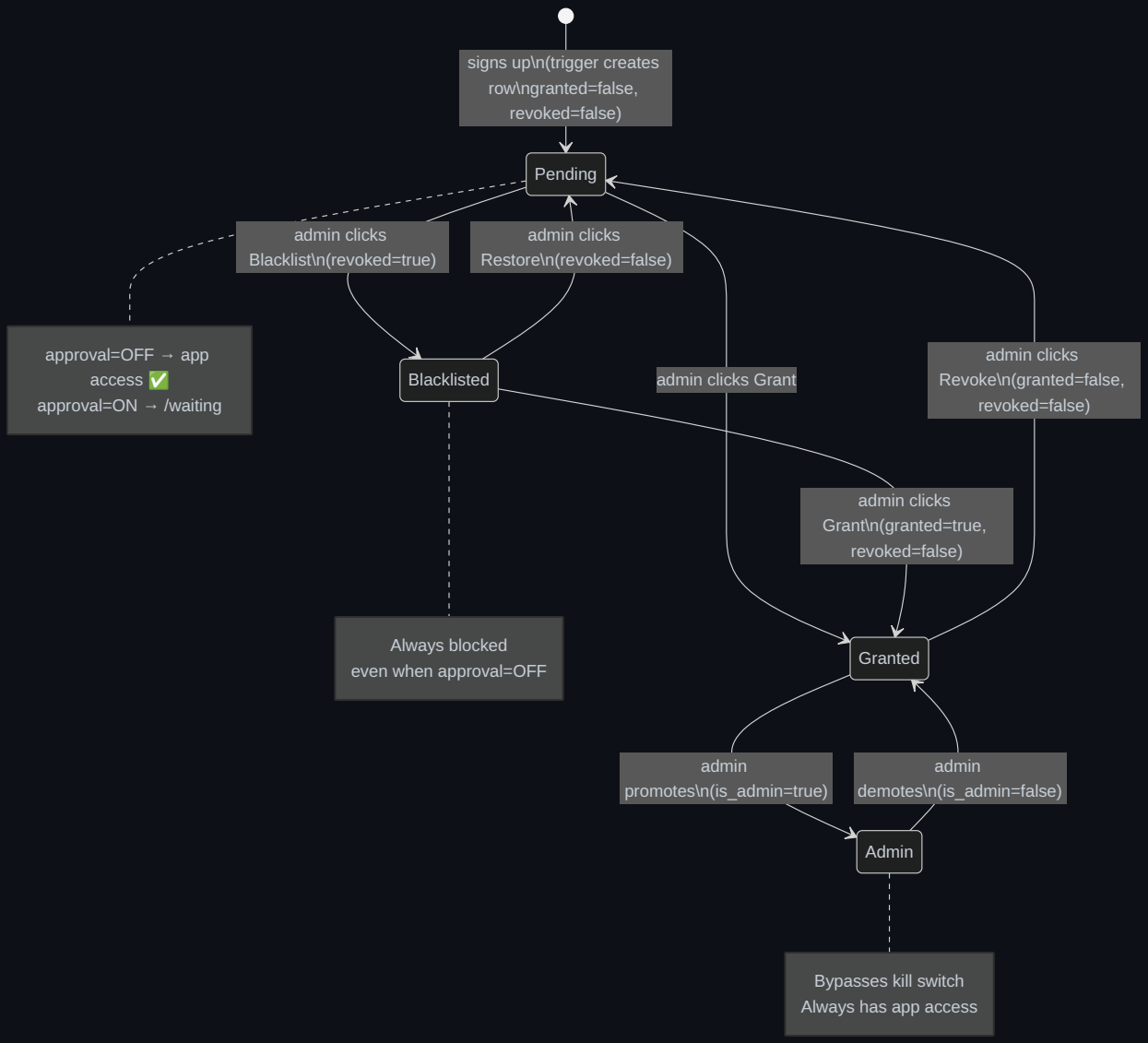
2 — Database Schema (ER)



3 — Middleware Decision Flow



4 — User Lifecycle State Machine



Overview

DSA Tracker is a personal, invite-only web application for tracking LeetCode and other DSA problem-solving progress. It generates personalised daily queues, tracks streaks and history, and provides an admin panel for user access management.

Stack

Layer	Technology
Framework	Next.js 16 (App Router)
Language	TypeScript 5
UI	React 19, Vanilla CSS
Database	Supabase (PostgreSQL)
Auth	Supabase Auth (Google OAuth + email/password)
Hosting	Vercel (serverless edge)

Layer	Technology
Math rendering	KaTeX

Key libraries:

- `@supabase/supabase-js` v2 — direct DB client (service role, admin operations)
- `@supabase/ssr` v0.12 — session-aware server/browser clients for Next.js

---

Repository Layout

```
dsa-tracker/
├── app/                    # Next.js App Router pages + API routes
│   ├── layout.tsx          # Root layout (fonts, global CSS)
│   ├── page.tsx            # Main queue page (/)
│   ├── admin/page.tsx      # Admin panel (/admin)
│   ├── analytics/page.tsx  # Analytics (/analytics)
│   ├── history/page.tsx    # Queue history (/history)
│   ├── problems/page.tsx   # Problem browser (/problems)
│   ├── settings/page.tsx   # User settings (/settings)
│   ├── login/page.tsx      # Auth page (/login)
│   ├── waiting/page.tsx    # Access pending / maintenance (/waiting)
│   ├── auth/callback/      # OAuth callback handler
│   └── api/
│       ├── queue/          # Queue generation + history
│       │   ├── route.ts    # GET (fetch today's queue), POST (regenerate)
│       │   ├── skip/       # POST (skip a problem)
│       │   └── history/    # GET (queue history with versions)
│       ├── progress/       # PATCH (update problem status)
│       ├── streak/         # GET (current streak)
│       ├── settings/       # GET + PATCH (user settings)
│       └── admin/
│           ├── users/      # GET + POST (list / pre-approve)
│           │   └── [userId]/ # PATCH + DELETE (grant/revoke/blacklist)
│           ├── stats/      # GET (per-user analytics)
│           └── settings/   # GET + PATCH (system toggles)
├── lib/
│   ├── supabase-server.ts  # Cookie-aware server client (API routes)
│   ├── supabase-browser.ts # Browser client (client components)
│   └── supabase-admin.ts   # Service-role client (admin operations)
├── middleware.ts            # Auth + access control on every request
├── components/             # Shared UI components
├── types/                  # Shared TypeScript types
├── config/                 # App configuration (topics, subtopics)
├── data/                   # Static data files
├── scripts/                # One-off scripts (data import, etc.)
└── docs/
    ├── migrations/         # All SQL migration files in order
    ├── plans/              # Feature plans (1.md … 6.1.md)
    └── reviews/            # Audits and reviews
```

Database Schema

All tables live in Supabase's `public` schema backed by PostgreSQL. `auth.users` is managed by Supabase Auth.

Core tables

`problems`

Seeded from LeetCode data. Read-only in production.

Column	Type	Notes
`id`	integer PK	LeetCode problem ID
`title`	text	
`title_slug`	text	URL slug
`difficulty`	text	`Easy` / `Medium` / `Hard`
`topic_tags`	text[]	e.g. `['array', 'dp']`
`description`	text	Problem statement
`ac_rate`	numeric	Acceptance rate
`paid_only`	boolean	
`olt_score`	numeric	Custom difficulty score
`sub_type`	text	e.g. `graph` , `tree`
`source`	text	`leetcode` / other
`external_url`	text	

`user_progress`

One row per user per problem. Tracks status and timing.

Column	Type	Notes
`user_id`	UUID FK → auth.users	
`problem_id`	integer FK → problems	
`status`	text	`TO_DO` / `ATTEMPTING` / `SOLVED`
`updated_at`	timestamptz	

Column	Type	Notes
`attempt_started_at`	timestamptz	When ATTEMPTING was first set
`solve_duration_secs`	integer	Seconds from start to SOLVED
`first_attempted_at`	timestamptz	

RLS: `ALL` — `user_id = auth.uid()`

`daily_queues`

One row per user per day. Stores the ordered list of problem IDs for the day.

Column	Type	Notes
`user_id`	UUID FK → auth.users	
`date`	date	Queue date
`problem_ids`	integer[]	Ordered list
`generated_at`	timestamptz	

RLS: `ALL` — `user_id = auth.uid()`

`queue_history`

Append-only log. Every regeneration creates a new row (version increments per day).

Column	Type	Notes
`id`	bigserial PK	
`user_id`	UUID FK → auth.users	
`date`	date	
`problem_ids`	integer[]	
`version`	integer	Increments on regen (1, 2, 3...)
`generated_at`	timestamptz	

RLS: `ALL` — `user_id = auth.uid()`

`user_settings`

Per-user queue configuration.

Column	Type	Default	Notes

Column	Type	Default	Notes
`user_id`	UUID PK → auth.users		
`daily_count`	integer	5	Total problems per day (1–20)
`hard_count`	integer	2	
`medium_count`	integer	3	
`easy_count`	integer	0	
`difficulty_mode`	text	`standard`	`chill` / `standard` / `grind` / `custom`
`enabled_topic_slugs`	text[]	null	null = all topics
`topic_weight_overrides`	jsonb	`{}`	`{ "dp": 2.0 }` = 2× weight
`enabled_sources`	text[]	`['leetcode']`	
`cooldown_days`	integer	7	Days before a solved problem re-enters pool

RLS: `ALL` — `user_id = auth.uid()`

`user_access`

Access control gate. One row per registered user.

Column	Type	Default	Notes
`user_id`	UUID PK → auth.users		
`granted`	boolean	false	Admin approved
`revoked`	boolean	false	Admin explicitly blacklisted
`is_admin`	boolean	false	Admin panel access
`email`	text	null	Cached from auth.users
`display_name`	text	null	Cached from OAuth metadata
`requested_at`	timestamptz	now()	When the row was created
`admin_notes`	text	null	Free-text notes by admin

RLS: `SELECT` only — `user_id = auth.uid()` (writes via service role only)

User lifecycle states:

State	`granted`	`revoked`	`is_admin`	Notes
Pending	false	false	false	Signed up, awaiting review
Granted	true	false	false	Approved
Revoked	false	false	false	Temporarily un-granted (back to Pending)
Blacklisted	any	**true**	false	Explicitly blocked
Admin	any	false	**true**	Always passes middleware
Orphaned	—	—	—	No row (trigger failure or deleted)

DB trigger `on_auth_user_access_created`: fires `AFTER INSERT ON auth.users`, inserts `(user_id, granted=false, email, display_name)` with `ON CONFLICT DO NOTHING`.

`system_settings`

Single-row global toggles. Enforced by `CHECK (id = 1)`.

Column	Type	Default	Notes
`id`	integer PK	1	Always 1
`approval_required`	boolean	true	Gate for new user access
`site_enabled`	boolean	true	Kill switch
`updated_at`	timestamptz	now()	

RLS: ON but no policies — service role key bypasses; anon key is blocked by default.

Views

`problems_with_status`

Joins `problems` with the current user's `user_progress` row via `auth.uid()`. Exposes `status`, `status_updated_at`, timing columns.

Created with `security_invoker = true` — RLS is applied using the caller's identity (not the view creator's), eliminating the Supabase Security Definer warning.

Authentication & Access Flow

Signup / Login

```
Browser → /login
  ├─ Google OAuth → Supabase → /auth/callback?code=…
  │     └─ exchangeCodeForSession()
  │         ├─ Sets session cookies on redirect response
  │         └─ Upsert user_access (ignoreDuplicates=true — preserves granted=
  │             └─ Redirect → /
  └─ Email+password → signUp() / signInWithPassword()
        └─ Email confirmation required before session is valid
```

Middleware (runs on every non-static request)

`middleware.ts` executes in the **Vercel Edge Runtime** on every request matching the config matcher (excludes `_next/static`, images, favicon).

Execution order:

```
1. Refresh session (supabase.auth.getUser())
2. Fetch system_settings (service role, single-row PK lookup, ~5ms)
3. /waiting smart block (logged-in users only):
     - Re-evaluate actual access state on every /waiting request
     - Redirect admins → /
     - Redirect granted/pending-with-approval-off → /
     - Fix stale ?reason= param if site came back up
4. Unauthenticated + non-public path → /login (or /waiting?reason=maintenance
5. Unauthenticated + /login + site disabled → /waiting?reason=maintenance
6. Kill switch (site_enabled=OFF) + logged-in non-admin → /waiting?reason=mai
7. Whitelist check (approval_required=ON):
     - Fetch user_access (granted, is_admin)
     - Not granted → /waiting
     - /admin path: re-verify is_admin via service role
8. Whitelist check (approval_required=OFF):
     - Fetch user_access (granted, is_admin, revoked)
     - revoked=true → /waiting (blacklisted, always blocked)
     - /admin path: re-verify is_admin via service role
```

Public paths (always pass through): `/login`, `/auth/callback`, `/waiting`

Safe fallback: any DB error in `getSystemSettings()` returns `{ approval_required: true, site_enabled: true }` — most restrictive, never accidentally opens.

API Routes

All routes are **Next.js Route Handlers** (server-side, Node.js runtime, not Edge).

Every route authenticates via `supabase.auth.getUser()` as first step. `user_id` is always taken from the server-verified JWT — never from the request body.

Queue

Method	Route	Description
`GET`	`/api/queue`	Fetch today's queue (generates if missing)
`POST`	`/api/queue`	Regenerate today's queue
`POST`	`/api/queue/skip`	Skip a problem, get replacement
`GET`	`/api/queue/history`	Queue history, grouped by date + version

Queue generation algorithm (`app/api/queue/route.ts`):

1. Load `user_settings` (daily_count, difficulty split, topic filters, cooldown_days, sources)
2. Fetch all problems matching source/topic filters
3. Exclude problems solved within `cooldown_days`
4. Weight by `topic_weight_overrides` and `olt_score`
5. Pick problems matching the difficulty split (hard/medium/easy counts)
6. Write to `daily_queues` , append to `queue_history` (version++)

Progress

Method	Route	Description
`PATCH`	`/api/progress`	Update a problem's status (TO_DO / ATTEMPTING / SOLVED)

Streak

Method	Route	Description
`GET`	`/api/streak`	Returns current streak (consecutive days with ≥1 SOLVED)

Settings

Method	Route	Description
`GET`	`/api/settings`	Fetch user settings
`PATCH`	`/api/settings`	Update user settings

Admin (all require `is_admin=true`)

Method	Route	Description
`GET`	`/api/admin/users`	List all users (with access rows + auth enrichment)
`POST`	`/api/admin/users`	Pre-grant by email
`PATCH`	`/api/admin/users/[userId]`	Grant / revoke / promote / notes
`DELETE`	`/api/admin/users/[userId]`	Blacklist (sets `revoked=true`, keeps row)
`GET`	`/api/admin/stats`	Per-user analytics (solved count, topics, last active)
`GET`	`/api/admin/settings`	Fetch system toggles
`PATCH`	`/api/admin/settings`	Update system toggles

Admin auth helper (`getAdminUser()`): reads session cookie → `getUser()` → fetches `user_access.is_admin` via service role client → returns user or null.

Supabase Clients

Three distinct client configurations, each used in the right context:

Client	File	Key used	Context
Server (session-aware)	`lib/supabase-server.ts`	`NEXT_PUBLIC_SUPABASE_ANON_KEY`	API routes, server components — respects RLS via user session
Browser	`lib/supabase-browser.ts`	`NEXT_PUBLIC_SUPABASE_ANON_KEY`	Client components (`'use client'`)
Admin (service role)	`lib/supabase-admin.ts`	`SUPABASE_SERVICE_ROLE_KEY`	Admin API routes, middleware — bypasses RLS

> **Caution**
>
> `SUPABASE_SERVICE_ROLE_KEY` has no `NEXT_PUBLIC_` prefix — Next.js never bundles it into client-side JavaScript. It is used exclusively in server-side code.

Pages

Route	Auth required	Notes
`/`	✅ Granted	Main queue view, streak, progress controls
`/problems`	✅ Granted	Full problem browser with filters
`/analytics`	✅ Granted	Personal stats (topics, difficulty split)
`/history`	✅ Granted	Queue history by date, expandable versions
`/settings`	✅ Granted	Queue config (count, difficulty, topics, sources)
`/admin`	✅ is_admin	User management + system toggles + analytics
`/login`	Public*	Google OAuth + email/password
`/waiting`	Public	Access pending OR maintenance (based on `?reason=`)
`/auth/callback`	Public	OAuth code exchange — always must be accessible

* `/login` redirects to `/waiting?reason=maintenance` when `site_enabled=OFF`.

`/waiting` page behaviour

Reads session on mount (`supabase.auth.getSession()`). Renders differently based on:

`?reason=`	`isLoggedIn`	Shows
(absent)	true	"Access Pending" + Sign Out
`maintenance`	true	"Down for Maintenance" + Sign Out
`maintenance`	false	"Down for Maintenance" (no sign out, no admin hint)

Access Control Matrix

When `approval_required=ON`:

User state	`/` and app	`/admin`	`/waiting`
Unauthenticated	→ /login	→ /login	✅ (public)

User state	/ and app	/admin	/waiting
Pending	→ /waiting	→ /waiting	✅
Blacklisted	→ /waiting	→ /waiting	✅
Granted	✅	→ /	✅ (redirected to /)
Admin	✅	✅	✅ (redirected to /)

When `approval_required=OFF`:

User state	/ and app	Notes
Pending / Orphaned	✅	Never explicitly blocked
Blacklisted (`revoked=true`)	→ /waiting	Always blocked regardless of toggle
Granted / Admin	✅	

When `site_enabled=OFF` (kill switch):

User state	Result
Unauthenticated	→ /waiting?reason=maintenance
Unauthenticated + /login	→ /waiting?reason=maintenance
Non-admin	→ /waiting?reason=maintenance
Admin	✅ Passes through

Deployment

- **Hosting**: Vercel (auto-deploy on push to `main`)
- **Environment variables**: set in Vercel project settings
  - `NEXT_PUBLIC_SUPABASE_URL`
  - `NEXT_PUBLIC_SUPABASE_ANON_KEY`
  - `SUPABASE_SERVICE_ROLE_KEY`
- **DB migrations**: run manually in Supabase SQL Editor (see `docs/migrations/` in chronological order)
- **Middleware runtime**: Vercel Edge (runs globally at CDN nodes)
- **API routes runtime**: Node.js serverless functions

Migration history (in order)

File	What it does

File	What it does
`phase2.sql`	`user_progress` timing cols, `user_settings`, view rebuild
`phase4.sql`	`queue_history`, `user_access`, `problems` sub_type/source cols
`phase4e.sql`	Extended problem schema
`phase4f.sql`	Minor schema fix
`admin_panel.sql`	`user_access` email/display_name cols, updated trigger
`fix_new_user_trigger.sql`	Rebuild trigger with exception guard
`subtopics_config.sql`	Subtopics config table
`system_settings.sql`	Global toggles table (Plan 5)
`plan6_fixes.sql`	Backfill null email/display_name; recreate view with security_invoker
`plan6.1_display_name.sql`	Fix display_name rows incorrectly set to email
`plan6.2_revoked_column.sql`	Add `revoked` boolean to `user_access`

Security Model

Concern	Mitigation
Unauthenticated API access	Every route calls `getUser()` → 401 if no session
Cross-user data reads	All queries scoped to `user.id` (server-verified JWT), RLS as defense-in-depth
Admin privilege escalation	`user_access` has SELECT-only RLS for regular users; is_admin checked via service role
Service role key exposure	No `NEXT_PUBLIC_` prefix; never in browser bundle
Bot account creation	Email confirmations required (Supabase setting, verified ON)
OAuth redirect hijack	`redirectTo` built from `window.location.origin`, validated by Supabase allow-list
`/waiting` dead-end stale state	Middleware re-evaluates every logged-in `/waiting` request and redirects correctly

Full audit: `docs/reviews/public_sharing_audit.md` — all checks passed Oct 13, 2026.